

From Validated Assembly to Moving-Cloud Renormalization

A Ten-Layer Audit and the Minimal Stage-A/A5 Completion Frontiers

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Abstract

RH-72–RH-80 carried the small-noise directional program from validated folded-Gaussian assembly through intrinsic square determinants and into the first relative fixed-disk construction. This tenth-layer review determines exactly which primitive gates remain, including the disjunctive structure hidden by a linear roadmap.

We prove a completion-antichain theorem for monotone AND/OR dependency formulas. Closed leaves contribute the empty completion, open leaves their singleton, OR takes the inclusion-minimal union of child families, and AND takes inclusion-minimal unions across their Cartesian product. The resulting family is exactly the set of inclusion-minimal open-gate bundles whose closure makes the target true.

For the archived program,

$$A_1 = F \wedge (L \vee E),$$

where the finite-scale chain F is now closed, L is the all-level log-square block law, and E is the all-level postblock effective-rank law. Thus Stage A has exactly two singleton completion alternatives: $\{L\}$ and $\{E\}$. The effective-rank corridor is preferred after the single-arc route failed and rank four captured at least 99.9999% at all anchors. Either corridor would also close unconditional intrinsic identification and shrinking-disk determinants by RH-78–RH-79.

The relative fixed-disk target has two minimal four-gate bundles:

$$\{L, P, C, U\}, \quad \{E, P, C, U\},$$

where P is an actual moving-cloud Riesz projection, C its coefficient bridge, and U a uniform trace-class complementary limit. Generic absolute continuity and fixed scalar pole cancellation are both closed false shortcuts, while moving-cloud factorization is green algebraically.

A 256-bit Arb audit certifies the archived route margins: at least 99.783% of the finite Hardy slack remains, the shrinking-disk determinant bound improves by a factor exceeding 100.37, the fixed-disk bound reverses by a factor exceeding 19.47, and the RH-80 inside/outside pole-fork contrast exceeds 5.19×10^8 . These certify the finite and model-level verdicts, not the open all-level or cloud-complement theorems. No Hilbert–Polya or Riemann-hypothesis conclusion is made.

Keywords: dependency frontier; monotone formula; validated numerics; effective rank; relative determinant; small noise.

MSC 2020: 47A55; 47B10; 37C30; 65G20; 68R10.

1 Scope and outcome

The previous checkpoint ended with two open Stage-A tasks: complete the finite upstream interval chain and prove uniform small-noise scaling. The first is now closed. RH-72 validated exact stochastic folded-Gaussian and Haar assembly [2]; RH-73 validated Perron/parity rank-two deflation [5]; and RH-74 joined these layers to the frozen terminal Hardy certificate at all five scales [4].

The uniform problem then split. RH-75 isolated a sufficient log-square block law [3]. RH-76 proved that a single narrow phase arc does not explain the observed horizons [1]. RH-77 found a stronger surviving finite-scale phenomenon: the postblock future is nearly finite rank [8]. RH-78 proved that an all-level theorem for either phenomenon closes Stage A with zero Hardy power and hence closes strict-mesh intrinsic identification [9].

RH-79 propagated that conditional result to trace-norm squares and shrinking-disk determinants, while locating the generic fixed-disk exponential wall [6]. RH-80 then proved that multiplication by the fixed limiting pole factor also fails across the cloud circle; the viable object is a quotient by the actual moving spectral cloud [7].

The outcome is not a linear list of five equally immediate tasks. It is an AND/OR network with two Stage-A alternatives followed by three simultaneous A5 gates. The purpose of this paper is to make that logic exact.

2 Minimal completion antichains

Definition 2.1 (Monotone dependency formula). A dependency formula is built recursively from primitive leaves using $\wedge$ and $\vee$. Each primitive leaf is currently either closed or open. A completion set is a set of open leaves which, when declared closed while all other statuses are held fixed, makes the formula true. It is minimal if no proper subset is a completion set.

For a finite family $\mathcal{F}$ of sets, let $\text{Min}(\mathcal{F})$ delete every member that strictly contains another member. Define $\mathfrak{M}(\Phi)$ recursively by

$$\begin{aligned} \mathfrak{M}(x) &= \{\emptyset\}, & x \text{ a closed leaf,} \\ \mathfrak{M}(x) &= \{\{x\}\}, & x \text{ an open leaf,} \\ \mathfrak{M}\left(\bigvee_i \Phi_i\right) &= \text{Min}\left(\bigcup_i \mathfrak{M}(\Phi_i)\right), \\ \mathfrak{M}\left(\bigwedge_i \Phi_i\right) &= \text{Min}\left\{\bigcup_i S_i : S_i \in \mathfrak{M}(\Phi_i)\right\}. \end{aligned}$$

Theorem 2.2 (Monotone completion-antichain theorem). *For every finite monotone dependency formula Φ , the family $\mathfrak{M}(\Phi)$ is exactly the set of inclusion-minimal completion sets of Φ . In particular, it is an antichain.*

Proof. Proceed by structural induction. The leaf statements are immediate. An OR formula becomes true exactly when at least one child becomes true, so its completion sets are the union of the child completion families; deleting supersets leaves precisely the minimal ones. An AND formula becomes true exactly when every child becomes true. One must therefore choose a completion for each child and close their union. Every completion arises this way, and again deleting supersets leaves exactly the inclusion-minimal choices. This proves the recursion and the antichain property. $\square$

The theorem is elementary but useful: it prevents a closed prerequisite from remaining on the active frontier, preserves genuine alternative corridors, and removes dominated route bundles automatically.

3 Application to Stage A and A5

Let

- F : the finite-scale assembly/deflation/Hardy chain;

- L : the all-level RH-75 log-square block law;
- E : the all-level RH-77 postblock effective-rank law;
- P : an actual moving-cloud Riesz projection;
- C : the actual cloud coefficient bridge;
- U : a uniform trace-class complementary limit.

The archived composition, square-transfer, and moving-cloud factorization theorems are closed in their conditional or algebraic scopes. The active formulas therefore reduce to

$$A_1 = F \wedge (L \vee E), \quad (1)$$

$$A_4 = A_1 \wedge (\text{closed conditional composition}), \quad (2)$$

$$A_{5,\text{rel}} = A_4 \wedge P \wedge C \wedge U \wedge (\text{closed moving-cloud algebra}). \quad (3)$$

Corollary 3.1 (Current completion frontiers). *The minimal completion families are*

$$\begin{aligned} \mathfrak{M}(A_1) &= \mathfrak{M}(A_4) = \{\{L\}, \{E\}\}, \\ \mathfrak{M}(A_{5,\text{rel}}) &= \{\{L, P, C, U\}, \{E, P, C, U\}\}. \end{aligned}$$

The unconditional shrinking-disk determinant has the same two singleton completion alternatives as Stage A.

Proof. RH-74 closes F . Substitute the current leaf statuses into Eqs. (1) to (3) and apply Theorem 2.2. $\square$

The corollary separates two meanings of “the next wall.” Stage A has one logical wall with two alternative theorem forms. A5 has three additional conjunctive operator gates after either Stage-A choice. Those A5 gates may be studied in parallel, but the relative fixed-disk target needs all three.

4 Nine-paper theorem ledger

The build script hashes and reads every RH-72–RH-80 summary. They contain forty-one boolean theorem gates, all true in their stated scopes. This is an archive consistency count, not a claim of forty-one independent mathematical theorems.

Paper	Route effect	Durable result	Remaining boundary
72	validated advance	folded-Gaussian assembly	peripheral factors
73	validated advance	Perron/parity rank-two deflation	Hardy propagation
74	finite closure	end-to-end five-scale bridge	all-level scaling
75	analytic corridor	sufficient log-square law	all-level proof
76	branch no-go	single-arc barrier	other compression routes
77	positive fallback	postblock rank compression	analytic rank decay
78	synthesis	either corridor closes Stage A	corridor premise
79	advance/barrier	square and shrinking-disk transfer	fixed disk
80	route correction	moving-cloud relative gate	actual cloud/complement

Table 1: The nine inputs to the RH-81 synthesis. “No-go” applies only to the named branch or proof method.

Three false shortcuts are now durably marked:

1. single-arc phase compression as the source of the production horizons;

2. generic absolute determinant continuity on a fixed small-noise disk;
3. fixed scalar double-pole cancellation across the finite-cloud circle.

None is a no-go theorem for the full program. Each one redirects the route to a surviving alternative already encoded in Corollary 3.1.

5 Validated route margins

The review audit uses Arb at 256-bit precision and exact binary64 rational lifting of the archived summary bounds. The six values in Table 2 serve different diagnostic roles and are not combined into one probability or theorem constant.

Certified metric	Lower bound	Interpretation
finite Hardy slack fraction remaining	0.99783070	RH-74 uses only about 0.217% of inherited slack
rank-two capture excess above 99%	8.45345×10^{-4}	every RH-77 channel clears the rank-two gate
rank-four excess above 99.9999%	9.20296×10^{-7}	every channel clears the six-nine target
shrinking-disk improvement factor	100.3731	RH-79 error decreases by over two orders of magnitude
fixed-disk reversal factor	19.4789	the standard bound worsens after its minimum
pole-fork inside/outside contrast	5.19408×10^8	RH-80 sharply separates interior convergence from exterior growth

Table 2: Outward-certified margins inherited by the route review.

The finite-scale chain is therefore not marginal, and the effective-rank corridor is quantitatively compelling at the anchors. Conversely, the two fixed-disk negative verdicts are not artifacts of tiny numerical differences. The open issue in each case is theorem-level uniformity or operator construction, not additional floating precision.

6 Revised route and next theorem

The preferred immediate target is an all-level postblock singular-value decay theorem. A representative sufficient form is

$$s_{r+1}(O_k A_k^{M_k} S_k) \leq C \text{polylog}(k) e^{-cr}, \quad (4)$$

or any estimate giving a $\text{polylog}(k)$ rank with zero power of σ_k . The proof should exploit smoothing after one production block, rank-two peripheral removal, and the low-dimensional source/observation geometry. RH-76 shows why compressing the raw phase measure into one arc is the wrong object.

If a rigorous lower bound defeats (4) with a forbidden noise power, the RH-75 full-block law remains the exact fallback. Stage A should be reassessed only if both singleton completion alternatives are ruled out under every admissible mesh schedule.

In parallel, A5 preparation should construct a moving contour and Riesz projection for the actual cloud, prove the cloud coefficient bridge, and control the complementary two-step trace norm. A5 should be reassessed only if every natural cloud extraction leaves a non-normal relative family on the required fixed domain.

The updated detailed roadmap is archived as `UPDATED_ROADMAP.md`. Later Stages B–D remain unopened. There is no canonical scattering completion, self-adjoint generator, intrinsic $T \log T$ counting law, prime-power trace formula, completed zeta identity, Hilbert–Polya operator, or Riemann Hypothesis result in this block.

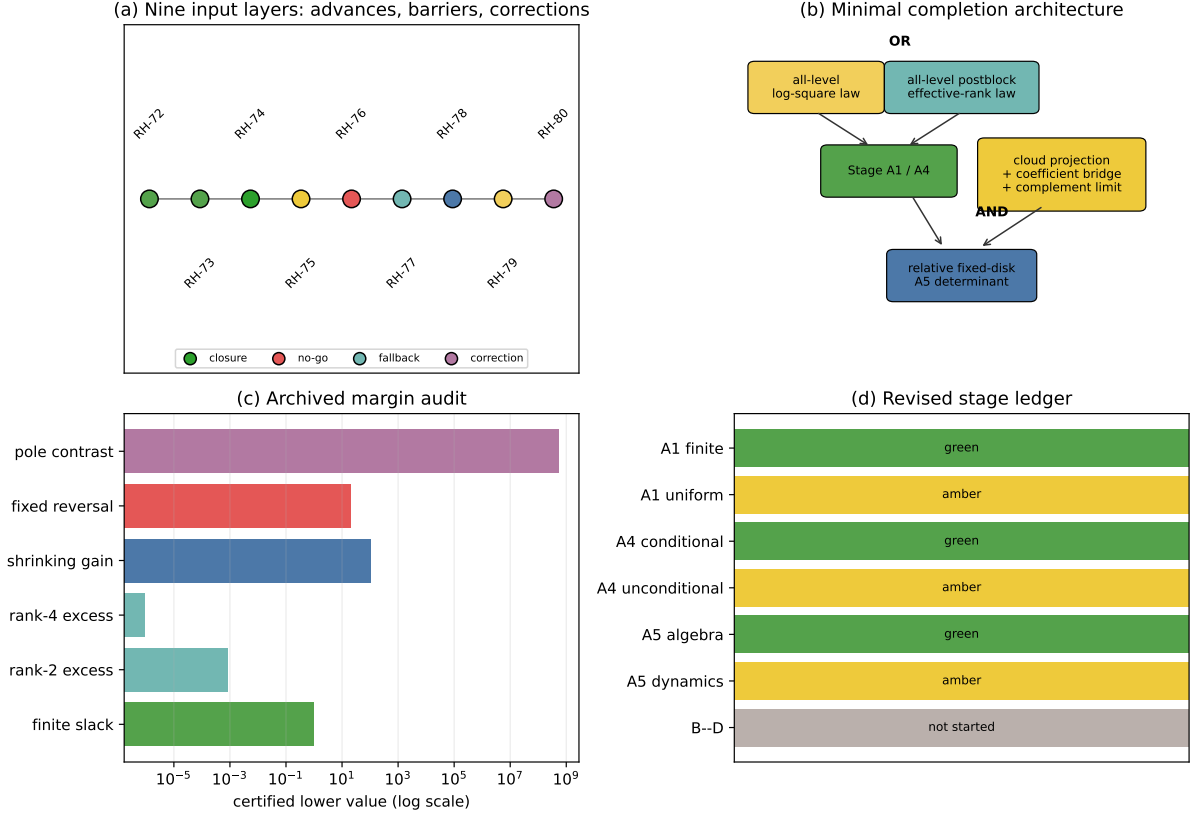


Figure 1: RH-72–RH-81 in one diagram: the input timeline, AND/OR completion architecture, certified margin audit, and revised stage ledger.

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